

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0876

Commonwealth Chesapeake 138 kV
March 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0876. The Interconnection Customer proposes a 120.1 MW solar photovoltaic facility interconnecting at the Commonwealth Chesapeake 138 kV substation in the Virginia area. The request was received on October 10, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 330 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,874,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Easton 2 345-138 kV Autotransformer
Contingency: Cady Substation-Easton 2 345.0 kV line
Loading: 192.9% (Rating: 871 MVA)
DFAX: 43.1%
Pre-project loading: 186.9%
2. Facility: Navy Mole-Easton 2 138.0 kV
Contingency: Easton 2-Cady 345.0 kV
Loading: 192.9% (Rating: 871 MVA)
DFAX: 43.1%
Pre-project loading: 186.9%
3. Facility: Chincoteague-Wattsville 138.0 kV
Contingency: Commonwealth Chesapeake-Southtown Substation 345.0 kV
Loading: 187.6% (Rating: 326 MVA)
DFAX: 24.0%
Pre-project loading: 178.7%
4. Facility: Ess Lwes-Rehoboth 138.0 kV Line
Contingency: Vienna Operations Substation-Easton 2 345.0 kV
Loading: 178.0% (Rating: 385 MVA)
DFAX: 25.1%
Pre-project loading: 170.2%
5. Facility: Saddlebrooke Ranch-Lammers 138.0 kV L
Contingency: Vienna Operations-Easton 2 345.0 kV
Loading: 171.7% (Rating: 752 MVA)
DFAX: 47.8%
Pre-project loading: 164.1%

6. Facility: Sollers Warf-Persimmon Gap 138.0 kV line
Contingency: Easton 2-Vienna Operations 345.0 kV
Loading: 171.3% (Rating: 547 MVA)
DFAX: 32.4%
Pre-project loading: 164.2%
7. Facility: Dodd Rd-Cady 345.0 kV
Contingency: Cady 345 kV/138 kV XFMR
Loading: 165.8% (Rating: 1,246 MVA)
DFAX: 60.9%
Pre-project loading: 159.9%
8. Facility: Rodessa-Hbbell 138.0 kV Line
Contingency: Southtown-Commonwealth Chesapeake 345.0 kV
Loading: 164.9% (Rating: 215 MVA)
DFAX: 12.4%
Pre-project loading: 158.0%
9. Facility: Henning-Kiwi 138.0 kV line
Contingency: Easton 2-Vienna Operations 345.0 kV
Loading: 162.6% (Rating: 657 MVA)
DFAX: 32.4%
Pre-project loading: 156.7%
10. Facility: Quince Orchard-Agawam 138.0 kV
Contingency: Commonwealth Chesapeake-Southtown 345.0 kV
Loading: 160.9% (Rating: 171 MVA)
DFAX: 9.6%
Pre-project loading: 154.1%
11. Facility: Vienna Operations 138-1 kV Autotransformer
Contingency: Easton 2-Vienna Operations 345.0 kV
Loading: 160.4% (Rating: 921 MVA)
DFAX: 47.6%
Pre-project loading: 154.2%
12. Facility: Hillsoro-Saltspring Island 138.0 kV Line
Contingency: Cady-Easton 2 345.0 kV
Loading: 158.9% (Rating: 242 MVA)
DFAX: 11.7%
Pre-project loading: 153.1%
13. Facility: Southtown-Vienna Operations 345.0 kV
Contingency: W -Station-St Bonifacius Substation 138.0 kV
Loading: 158.6% (Rating: 2,011 MVA)
DFAX: 88.1%
Pre-project loading: 153.3%
14. Facility: Commonwealth Chesapeake-Rodessa 138.0 kV Line
Contingency: Southtown-Commonwealth Chesapeake 345.0 kV
Loading: 156.8% (Rating: 232 MVA)
DFAX: 12.4%
Pre-project loading: 150.4%
15. Facility: Shingle Crk-W -Station 138.0 kV L
Contingency: Southtown-Vienna Operations 345.0 kV
Loading: 156.7% (Rating: 906 MVA)
DFAX: 53.6%
Pre-project loading: 149.6%
16. Facility: Banta Carbona-Willow Pass 138.0 kV
Contingency: Dodd Road-Cady 345.0 kV

- Loading: 156.6% (Rating: 259 MVA)
DFAX: 12.0%
Pre-project loading: 151.0%
17. Facility: Southtown 345 kV/138 kV XFMR
Contingency: Southtown-Vienna Operations 345.0 kV
Loading: 156.1% (Rating: 1,078 MVA)
DFAX: 38.2%
Pre-project loading: 151.9%
18. Facility: Saltspring Island-Key Largo 138.0 kV L
Contingency: Cady-Easton 2 345.0 kV
Loading: 156.1% (Rating: 243 MVA)
DFAX: 11.7%
Pre-project loading: 150.3%
19. Facility: Ripple 345/138kV TX
Contingency: Vienna Operations-Southtown 345.0 kV line
Loading: 154.6% (Rating: 624 MVA)
DFAX: 35.9%
Pre-project loading: 147.7%
20. Facility: Shingle Creek-Southtown 138.0 kV
Contingency: Southtown-Vienna Operations 345.0 kV
Loading: 154.3% (Rating: 940 MVA)
DFAX: 53.6%
Pre-project loading: 147.4%
21. Facility: Memphis Junction-Lammers 138.0 kV L
Contingency: Easton 2-Vienna Operations 345.0 kV
Loading: 153.5% (Rating: 637 MVA)
DFAX: 32.4%
Pre-project loading: 147.4%
22. Facility: Quince Orchard-Southtown 138.0 kV Line
Contingency: Southtown-Commonwealth Chesapeake 345.0 kV
Loading: 151.0% (Rating: 336 MVA)
DFAX: 19.3%
Pre-project loading: 144.1%
23. Facility: Vienna Operations 345/1kV Auto
Contingency: Easton 2-Vienna Operations 345.0 kV Line
Loading: 150.6% (Rating: 981 MVA)
DFAX: 47.6%
Pre-project loading: 144.8%
24. Facility: Easton 2 Substation-Vienna Operations 345.0 kV
Contingency: Linde-Ripple 138.0 kV
Loading: 150.0% (Rating: 1,710 MVA)
DFAX: 76.8%
Pre-project loading: 144.6%
25. Facility: North Lebanon-Fairmont 138.0 kV Circuit 2
Contingency: Dodd Rad Substation-Cady 345.0 kV
Loading: 149.2% (Rating: 316 MVA)
DFAX: 14.8%
Pre-project loading: 143.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Easton 2 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$13,764,000	4% (\$549,000)
Navy Mole-Easton 2 138.0 kV	Rebuild line with higher-capacity conductor	\$551,000	4% (\$22,000)
Chincoteague-Watts ville 138.0 kV	Rebuild line with higher-capacity conductor	\$7,967,000	5% (\$377,000)
Ess Lewes-Rehoboth 138.0 kV	Rebuild line with higher-capacity conductor	\$3,606,000	4% (\$140,000)
Saddlebrooke Ranch Substation-Lammers 138.0 kV Line	Rebuild line with higher-capacity conductor	\$7,622,000	4% (\$285,000)
Sollers Warf-Persimmon Gap 138.0 kV line	Rebuild line with higher-capacity conductor	\$15,467,000	4% (\$552,000)
Dodd Rd-Cady 345.0 kV	Rebuild line with higher-capacity conductor	\$58,353,000	4% (\$2,215,000)
Hubbell-Rodessa 138.0 kV	Rebuild line with higher-capacity conductor	\$3,386,000	5% (\$176,000)
Henning-Kiwi 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,264,000	4% (\$134,000)
Agawam-Quince Orchard 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,035,000	6% (\$122,000)
Vienna Operations 138-1 kV Autotransformer	Replace transformer with higher-rated unit	\$4,160,000	4% (\$179,000)
Hillsoro-Saltspring Island 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,056,000	4% (\$185,000)
Southtown-Vienna Operations 345.0 kV	Rebuild line with higher-capacity conductor	\$38,852,000	4% (\$1,513,000)
Commonwealth Chesapeake-Rodessa 138.0 kV Line	Rebuild line with higher-capacity conductor	\$7,678,000	5% (\$398,000)
Shingle Crk-W -Station 138.0 kV L	Rebuild line with higher-capacity conductor	\$11,141,000	4% (\$483,000)
Banta Carbona-Willow Pass 138.0 kV	Rebuild line with higher-capacity conductor	\$23,978,000	3% (\$821,000)
Southtown 345/138kV Transformer Bank	Replace transformer with higher-rated unit	\$14,093,000	4% (\$494,000)
Key Largo-Satspring Island 138.0 kV line	Rebuild line with higher-capacity conductor	\$25,049,000	4% (\$917,000)
Riple 345/138 kV XFMR	Replace transformer with higher-rated unit	\$14,056,000	4% (\$604,000)
Shingle Creek-Southtown 138.0 kV	Rebuild line with higher-capacity conductor	\$4,261,000	4% (\$185,000)
Memphis Junction-Lammers 138.0 kV L	Rebuild line with higher-capacity conductor	\$20,537,000	4% (\$732,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Quince Orchard-Southtown 138.0 kV	Rebuild line with higher-capacity conductor	\$2,746,000	6% (\$152,000)
Vienna Operations 345/1kV Auto	Replace transformer with higher-rated unit	\$19,869,000	4% (\$855,000)
Easton 2 Substation-Vienna Operations 345.0 kV	Rebuild line with higher-capacity conductor	\$64,982,000	4% (\$2,607,000)
N Lebanon-Fairmont 138.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$4,860,000	4% (\$177,000)
	Total allocated to project		\$14,874,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.